

Deep Learning Project 2026 Spring

Introduction

Incremental learning, also known as continuous learning or lifelong learning, aims to enable machines to acquire knowledge incrementally. Similar to humans, this pattern addresses the need for models to adapt to changing tasks without retraining. Class-incremental learning (CIL) is a typically challenging scenario in incremental learning, allowing models to acquire knowledge from separately arriving data categories.

However, incremental learning faces the challenge of catastrophic forgetting [1], where models experience significant performance degradation on previous tasks when learning new ones. For example, in the classification task shown in Figure 1, after learning new tasks involving tigers and fish, the model misclassifies previous task images of birds as fish.

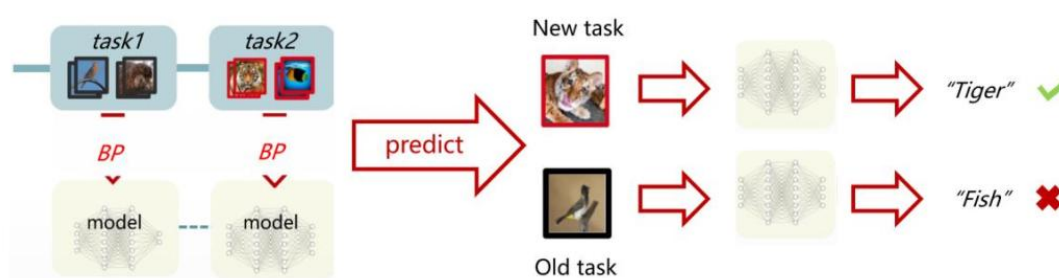


Figure 1. An example of catastrophic forgetting.

Several current approaches to incremental learning generally train a pre-trained model with certain knowledge storage on new data. The process of equipping the model with certain knowledge storage is typically referred to as the base training. In experimental validation, half of the dataset is commonly used during this base training, and the base training is crucial for the subsequent incremental learning to alleviate the catastrophic forgetting. In this project, you are required to refine the base training and improve the model's performance in the subsequent incremental learning stage.

Preliminary

1. Training agenda in class-incremental learning

In class incremental learning, we can assume that the model has obtained some basic knowledge during the base training. The base training in CIL generally uses half categories in the dataset and the other half is for the incremental learning. The base training is also called

Phase #0. The incremental learning is conducted in a phase-wise manner. That is, the remaining categories are evenly divided into several phases, then the model first trained in the base training is trained on the following separated data in sequence.

2. Analytic class-incremental learning

Analytic class-incremental learning (ACIL) is a recently proposed method for the CIL problem in an analytic manner [2]. This method fixes the feature extractor during the incremental learning and provides identical result to that obtained by training with all the data jointly. This property enables ACIL to be good indicator of backbone performance on the incremental learning.

Task

Use all the methods and tricks, including **you learned in the lectures or from other resources**, to enhance the backbone and finally obtain better performance in the incremental learning. You should follow these requirements.

1. **You can only use first half of the dataset to enhance the backbone.** The usage of the dataset during incremental learning will lead to data leakage and unconvincing results.
2. **This project is a team-work.** You should find a group with 3-5 students and work together.
3. **Use ACIL as the benchmark for incremental learning.** You should use another half categories to do the incremental learning with ACIL for fair comparison.
4. **The UC Merced Land Use dataset is recommended in this task.**
5. **The model architecture is recommended to be ViT-B_16.** Since this project is to enhance the backbone, directly using another backbone with more parameters can be effective but less elegant. Also, a backbone with fewer parameters has faster training speed.
6. **A project report with 5 pages and an oral presentation with slides should be completed.** The report and slides should be in English and you should write the report with the template. The oral presentation can be in Chinese.
7. **Grading Metric.** The project accounts for 60% of the total grade and the grading will consider the following 5 parts.
 - **Performance (20%).** The performance will be evaluated not only by the test accuracy but also the model efficiency and contents in the presentation/report.
 - **Workload (20%).** The individual workload of students in the course project will be

assessed based on the report and presentation, including attempts with different algorithms, data managements, analysis of various algorithms, and others.

- **Novelty (20%).** The innovation of the algorithms will be evaluated whether only common existing methods were used or whether innovative approaches were attempted (the results do not need to be highly effective, as long as they are reasonable)
- **Presentation (20%).** Presentation skills (including PPT and oral presentation) will be assessed. Clarity, standardization, and the quality of presentation materials will be focused.
- **Report (20%).** The course project report is a key component of the final project, aiming to evaluate the groups' mastery of the course knowledge as well as their ability to present the project effectively

Submission

Please submit the report and presentation slides to wiyanyue@mail.scut.edu.cn. The submission deadline is **00:00, June 16th (Week 16, Tuesday)**. The time for oral presentation is **19:00 pm, June 16th (Week 16, Tuesday)** and the location will be released quite soon.

Materials may help

1. Github repository of ACIL: <https://github.com/ZHUANGHP/Analytic-continual-learning>
2. Some blogs and papers:
 - (1) Introduction to incremental learning: <https://zhuanlan.zhihu.com/p/438766442>.
 - (2) ACIL paper:
https://proceedings.neurips.cc/paper_files/paper/2022/hash/4b74a42fc81fc7ee252f6bcb6e26c8be-Abstract-Conference.html.
 - (3) Chinese tutorial of ACIL:
https://www.bilibili.com/video/BV1wq421A7YM/?spm_id_from=333.999.0.0&vd_source=328610809c72f2a6f59b352853af9774.
 - (4) Potential solutions: label augmentation and self-supervised learning for backbone enhancing:
<https://proceedings.mlr.press/v119/lee20c.html>.

https://openaccess.thecvf.com/content/CVPR2021/html/Zhu_Prototype_Augmentation_and_Self-Supervision_for_Incremental_Learning_CVPR_2021_paper.html.

Reference

- [1] G. M. Van de Ven, T. Tuytelaars, and A. S. Tolias. "Three types of incremental learning", *Nature Machine Intelligence*, 4(12), 1185-1197, 2022.
- [2] H. Zhuang, Z. Weng, H. Wei, et al. "ACIL: Analytic Class Incremental Learning with Absolute Memorization and Privacy Protection," in *Advances in Neural Information Processing Systems*, 2022

Warm tips:

1. The training of model via backpropagation is **time-consuming**, it is recommended to begin the project in advanced.
2. Please do not tune the parameters blindly. You can seek existing experiences for hyperparameters to save time. You can also search for some advanced methods. Sometimes, using a better technique, such as label augmentation, can be effective than tedious parameter tuning.
3. It is better to know how the ACIL codes work. The backbone of your own should be loaded by the ACIL codes to check the performance.
4. If you have any questions, please contact the TA.